LAB 6

Implementation of Various Join Operations

Aim: *Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.*

1 [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

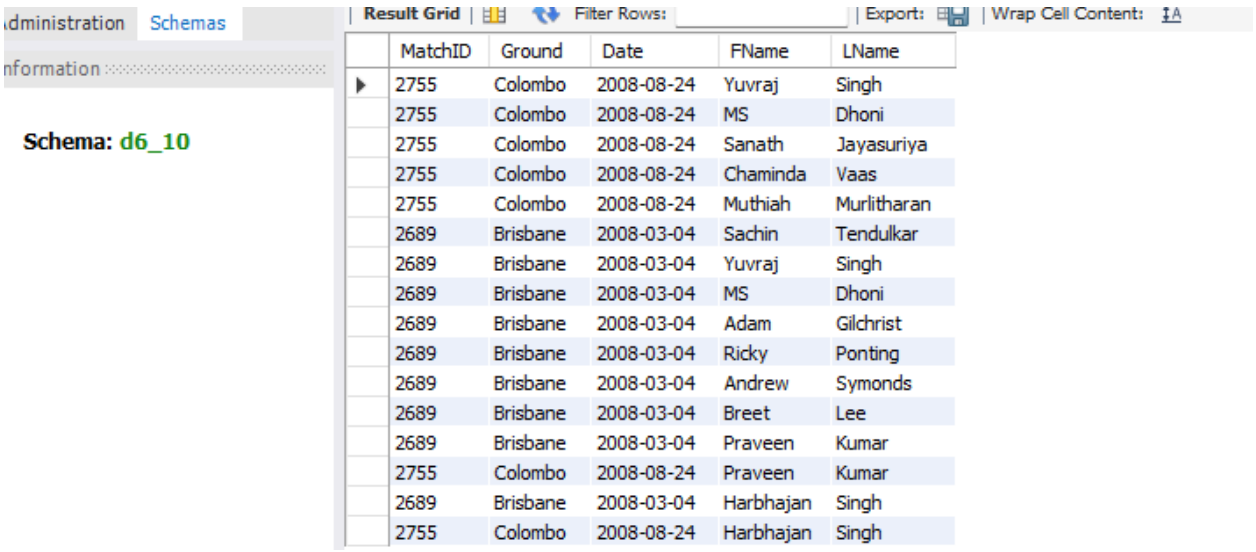
```
USE d6_10;

SELECT M.MatchID, M.Ground, M.Date, P.FName, P.LName

FROM ODIMatch M

INNER JOIN Batting B ON M.MatchID = B.MatchID

INNER JOIN Player P ON B.PID = P.PlayerID;
```



MatchID	Ground	Date	FName	LName
2755	Colombo	2008-08-24	Yuvraj	Singh
2755	Colombo	2008-08-24	MS	Dhoni
2755	Colombo	2008-08-24	Sanath	Jayasuriya
2755	Colombo	2008-08-24	Chaminda	Vaas
2755	Colombo	2008-08-24	Muthiah	Murlitharan
2689	Brisbane	2008-03-04	Sachin	Tendulkar
2689	Brisbane	2008-03-04	Yuvraj	Singh
2689	Brisbane	2008-03-04	MS	Dhoni
2689	Brisbane	2008-03-04	Adam	Gilchrist
2689	Brisbane	2008-03-04	Ricky	Ponting
2689	Brisbane	2008-03-04	Andrew	Symonds
2689	Brisbane	2008-03-04	Breet	Lee
2689	Brisbane	2008-03-04	Praveen	Kumar
2755	Colombo	2008-08-24	Praveen	Kumar
2689	Brisbane	2008-03-04	Harbhajan	Singh
2755	Colombo	2008-08-24	Harbhajan	Singh

2 [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

```
SELECT P.FName, P.LName, P.Country, B.NRuns, B.Six

FROM Player P

INNER JOIN Batting B ON P.PlayerID = B.PID
```

WHERE B.Six > 0;

Schema: d6_10

FName	LName	Country	NRuns	Six
Sanath	Jayasuriya	Sri Lanka	60	2
Yuvraj	Singh	India	38	2
MS	Dhoni	India	36	1
Andrew	Symonds	Australia	42	1

3 [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

SELECT P.PlayerID, P.FName, P.LName, B.MatchID, B.NRuns

FROM Player P

LEFT OUTER JOIN Batting B ON P.PlayerID = B.PID;

Schema: d6_10

PlayerID	FName	LName	MatchID	NRuns
21001	Virender	Sehwag	NULL	NULL
23001	Yuvraj	Singh	2689	38
23001	Yuvraj	Singh	2755	0
24001	Andrew	Symonds	2689	42
25001	MS	Dhoni	2689	36
25001	MS	Dhoni	2755	71
27001	Praveen	Kumar	2755	2
27001	Praveen	Kumar	2689	7
27002	Ishant	Sharma	NULL	NULL
89001	Sachin	Tendulkar	2689	91
90001	Brain	Lara	NULL	NULL
90002	Anil	Kumble	NULL	NULL
91001	Sanath	Jayasuriya	2755	60

Schema: d6_10

Result Grid | Filter Rows: | Exp

PlayerID	FName	LName	MatchID	NRuns
91001	Sanath	Jayasuriya	2755	60
92001	Shane	Warne	NULL	NULL
92002	Muthiah	Murlitharan	2755	1
92003	Inzamam	UI-Huq	NULL	NULL
93002	Heath	Streak	NULL	NULL
93003	Gary	Kirsten	NULL	NULL
94002	Chaminda	Vaas	2755	17
94004	Stephen	Fleming	NULL	NULL
95001	Ricky	Ponting	2689	1
95002	Shaun	Pollock	NULL	NULL
95003	Jacques	Kallis	NULL	NULL
96001	Rahul	Dravid	NULL	NULL
96002	Herschelle	Gibbs	NULL	NULL

Schema: d6_10

PlayerID	FName	LName	MatchID	NRuns
95003	Jacques	Kallis	NULL	NULL
96001	Rahul	Dravid	NULL	NULL
96002	Herschelle	Gibbs	NULL	NULL
96003	Saurav	Ganguly	NULL	NULL
97004	Daniel	Vettori	NULL	NULL
98001	Shahid	Afridi	NULL	NULL
98002	Harbha...	Singh	2755	2
98002	Harbha...	Singh	2689	3
99001	Breet	Lee	2689	7
99002	Adam	Gilchrist	2689	2
99003	Michale	Vaughan	NULL	NULL
300311	Michale	Clark	NULL	NULL
406317	Sam	NULL	NULL	NULL

4 [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

```
SELECT B.MatchID, B.NOvers, B.Nwickets,
    P.FName, P.LName, P.Country
FROM Player P
RIGHT OUTER JOIN Bowling B ON P.PlayerID = B.PID;
```

Schema: d6_10

MatchID	NOvers	Nwickets	FName	LName	Country
2689	10	1	Breet	Lee	Australia
2689	3	1	Andrew	Symonds	Australia
2689	3	0	Yuvraj	Singh	India
2755	9	1	Chaminda	Vaas	Sri Lanka
2755	10	1	Muthiah	Murlitharan	Sri Lanka
2755	4	0	Sanath	Jayasuriya	Sri Lanka
2755	10	2	Yuvraj	Singh	India
2755	10	3	Harbhajan	Singh	India
2689	10	1	Harbhajan	Singh	India

5 [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

```
SELECT B.MatchID, B.PID, B.NRuns AS BattingRuns, B.Six,
```

```
    W.NOvers, W.Nruns AS BowlingRuns, W.Nwickets
FROM Batting B
LEFT JOIN Bowling W
  ON B.MatchID = W.MatchID AND B.PID = W.PID
UNION
SELECT W.MatchID, W.PID, B.NRuns AS BattingRuns, B.Six,
    W.NOvers, W.Nruns AS BowlingRuns, W.Nwickets
FROM Batting B
RIGHT JOIN Bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID;
```

Schema: d6_10

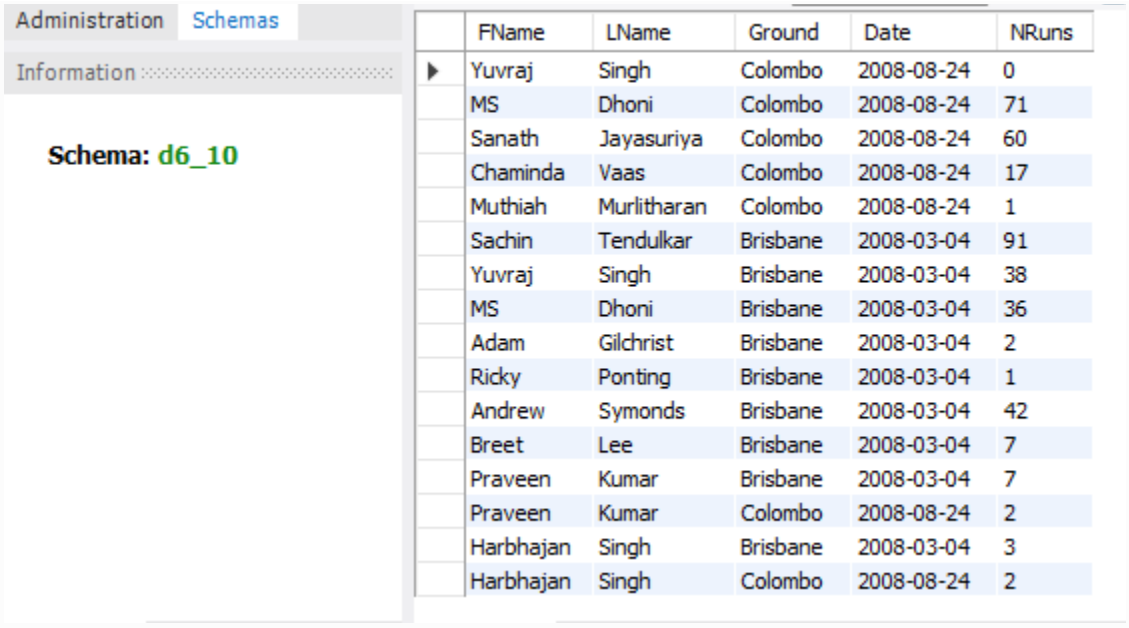
MatchID	PID	BattingRuns	Six	NOvers	BowlingRuns	Nwickets
2755	23001	0	0	10	53	2
2755	25001	71	0	NULL	NULL	NULL
2755	91001	60	2	4	29	0
2755	94002	17	0	9	40	1
2755	92002	1	0	10	56	1
2689	89001	91	0	NULL	NULL	NULL
2689	23001	38	2	3	15	0
2689	25001	36	1	NULL	NULL	NULL
2689	99002	2	0	NULL	NULL	NULL
2689	95001	1	0	NULL	NULL	NULL
2689	24001	42	1	3	27	1
2689	99001	7	0	10	58	1
2689	27001	7	0	NULL	NULL	NULL

Schema: d6_10

MatchID	PID	BattingRuns	Six	NOvers	BowlingRuns	Nwickets
2755	94002	17	0	9	40	1
2755	92002	1	0	10	56	1
2689	89001	91	0	NULL	NULL	NULL
2689	23001	38	2	3	15	0
2689	25001	36	1	NULL	NULL	NULL
2689	99002	2	0	NULL	NULL	NULL
2689	95001	1	0	NULL	NULL	NULL
2689	24001	42	1	3	27	1
2689	99001	7	0	10	58	1
2689	27001	7	0	NULL	NULL	NULL
2755	27001	2	0	NULL	NULL	NULL
2689	98002	3	0	10	44	1
2755	98002	2	0	10	40	3

6 [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

```
SELECT P.FName, P.LName, M.Ground, M.Date, B.NRuns
FROM Player P
INNER JOIN Batting B ON P.PlayerID = B.PID
INNER JOIN ODIMatch M ON B.MatchID = M.MatchID
WHERE M.Team1 = 'India' OR M.Team2 = 'India';
```



FName	LName	Ground	Date	NRuns
Yuvraj	Singh	Colombo	2008-08-24	0
MS	Dhoni	Colombo	2008-08-24	71
Sanath	Jayasuriya	Colombo	2008-08-24	60
Chaminda	Vaas	Colombo	2008-08-24	17
Muthiah	Murlitharan	Colombo	2008-08-24	1
Sachin	Tendulkar	Brisbane	2008-03-04	91
Yuvraj	Singh	Brisbane	2008-03-04	38
MS	Dhoni	Brisbane	2008-03-04	36
Adam	Gilchrist	Brisbane	2008-03-04	2
Ricky	Ponting	Brisbane	2008-03-04	1
Andrew	Symonds	Brisbane	2008-03-04	42
Breet	Lee	Brisbane	2008-03-04	7
Praveen	Kumar	Brisbane	2008-03-04	7
Praveen	Kumar	Colombo	2008-08-24	2
Harbhajan	Singh	Brisbane	2008-03-04	3
Harbhajan	Singh	Colombo	2008-08-24	2

7 [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

```
SELECT P.FName, P.LName, M.Ground, M.Date,
    B.NRuns, W.Nwickets
FROM Player P
INNER JOIN Batting B
  ON P.PlayerID = B.PID
```

```
INNER JOIN Bowling W
   ON B.MatchID = W.MatchID AND B.PID = W.PID
INNER JOIN ODIMatch M
ON B.MatchID = M.MatchID;
```

hema: d6_10

FName	LName	Ground	Date	NRuns	Nwickets
Yuvraj	Singh	Colombo	2008-08-24	0	2
Sanath	Jayasuriya	Colombo	2008-08-24	60	0
Chaminda	Vaas	Colombo	2008-08-24	17	1
Muthiah	Murlitharan	Colombo	2008-08-24	1	1
Yuvraj	Singh	Brisbane	2008-03-04	38	0
Andrew	Symonds	Brisbane	2008-03-04	42	1
Breet	Lee	Brisbane	2008-03-04	7	1
Harbhajan	Singh	Brisbane	2008-03-04	3	1
Harbhajan	Singh	Colombo	2008-08-24	2	3

8 [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

```
SELECT CONCAT(P1.FName, ' ', P1.LName) AS Player1_Name,
    CONCAT(P2.FName, ' ', P2.LName) AS Player2_Name,
    P1.Bplace
FROM Player P1
INNER JOIN Player P2
   ON P1.Bplace = P2.Bplace
AND P1.PlayerID < P2.PlayerID
```

Schema: d6_10

Player1_Name	Player2_Name	Bplace
Virender Sehwag	Ishant Sharma	Delhi
Gary Kirsten	Jacques Kallis	Cape Town
Gary Kirsten	Herschelle Gibbs	Cape Town
Jacques Kallis	Herschelle Gibbs	Cape Town

9 [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

```
SELECT M1.MatchID AS Match1_ID,
    M1.Ground,
    M1.Date AS Match1_Date,
    M2.MatchID AS Match2_ID,
    M2.Date AS Match2_Date
FROM ODIMatch M1
INNER JOIN ODIMatch M2
  ON M1.Ground = M2.Ground
 AND M1.Date <> M2.Date
AND M1.MatchID < M2.MatchID;
```

Schema: d6_10

Match1_ID	Ground	Match1_Date	Match2_ID	Match2_Date
2404	Colombo	2006-08-16	2750	2008-08-24
2404	Colombo	2006-08-16	2755	2008-08-24

10 [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

```
SELECT P.PlayerID, P.FName, P.LName,
    M.MatchID, M.Ground, M.Date
FROM Player P
CROSS JOIN ODIMatch M
WHERE P.Country = 'India';
```

Schema: d6_10

PlayerID	FName	LName	MatchID	Ground	Date
406317	Sam	NULL	2324	Peshwar	2006-02-06
98002	Harbhajan	Singh	2324	Peshwar	2006-02-06
96003	Saurav	Ganguly	2324	Peshwar	2006-02-06
96001	Rahul	Dravid	2324	Peshwar	2006-02-06
90002	Anil	Kumble	2324	Peshwar	2006-02-06
89001	Sachin	Tendulkar	2324	Peshwar	2006-02-06
27002	Ishant	Sharma	2324	Peshwar	2006-02-06
27001	Praveen	Kumar	2324	Peshwar	2006-02-06
25001	MS	Dhoni	2324	Peshwar	2006-02-06
23001	Yuvraj	Singh	2324	Peshwar	2006-02-06
21001	Virender	Sehwag	2324	Peshwar	2006-02-06
406317	Sam	NULL	2327	Rawalpi...	2006-02-11
98002	Harbhajan	Singh	2327	Rawalpi...	2006-02-11

Anil	Kumble	2404	Colombo	2006-08-16
Sachin	Tendulkar	2404	Colombo	2006-08-16
Ishant	Sharma	2404	Colombo	2006-08-16
Praveen	Kumar	2404	Colombo	2006-08-16
MS	Dhoni	2404	Colombo	2006-08-16
Yuvraj	Singh	2404	Colombo	2006-08-16
Virender	Sehwag	2404	Colombo	2006-08-16
Harbhajan	singh	2440	Mohali	2006-10-29
Saurav	Ganguly	2440	Mohali	2006-10-29
Rahul	Dravid	2440	Mohali	2006-10-29
Anil	Kumble	2440	Mohali	2006-10-29
Sachin	Tendulkar	2440	Mohali	2006-10-29
Ishant	Sharma	2440	Mohali	2006-10-29
Praveen	Kumar	2440	Mohali	2006-10-29
MS	Dhoni	2440	Mohali	2006-10-29
Yuvraj	Singh	2440	Mohali	2006-10-29
Virender	Sehwag	2440	Mohali	2006-10-29
Harbhajan	singh	2449	Cape T...	2006-11-26
Saurav	Ganguly	2449	Cape T...	2006-11-26
Rahul	Dravid	2449	Cape T...	2006-11-26
Anil	Kumble	2449	Cape T...	2006-11-26

Sachin	Tendulkar	2449	Cape T...	2006-11-26
Ishant	Sharma	2449	Cape T...	2006-11-26
Praveen	Kumar	2449	Cape T...	2006-11-26
MS	Dhoni	2449	Cape T...	2006-11-26
Yuvraj	Singh	2449	Cape T...	2006-11-26
Virender	Sehwag	2449	Cape T...	2006-11-26
Harbhajan	singh	2480	Nagpur	2007-01-21
Saurav	Ganguly	2480	Nagpur	2007-01-21
Rahul	Dravid	2480	Nagpur	2007-01-21
Anil	Kumble	2480	Nagpur	2007-01-21
Sachin	Tendulkar	2480	Nagpur	2007-01-21
Ishant	Sharma	2480	Nagpur	2007-01-21
Praveen	Kumar	2480	Nagpur	2007-01-21
MS	Dhoni	2480	Nagpur	2007-01-21
Yuvraj	Singh	2480	Nagpur	2007-01-21
Virender	Sehwag	2480	Nagpur	2007-01-21
Harbhajan	singh	2493	Vadodara	2007-01-31
Saurav	Ganguly	2493	Vadodara	2007-01-31
Rahul	Dravid	2493	Vadodara	2007-01-31
Anil	Kumble	2493	Vadodara	2007-01-31
Sachin	Tendulkar	2493	Vadodara	2007-01-31

Ishant	Sharma	2493	Vadodara	2007-01-31
Praveen	Kumar	2493	Vadodara	2007-01-31
MS	Dhoni	2493	Vadodara	2007-01-31
Yuvraj	Singh	2493	Vadodara	2007-01-31
Virender	Sehwag	2493	Vadodara	2007-01-31
Harbhajan	singh	2520	Rajkot	2007-02-11
Saurav	Ganguly	2520	Rajkot	2007-02-11
Rahul	Dravid	2520	Rajkot	2007-02-11
Anil	Kumble	2520	Rajkot	2007-02-11
Sachin	Tendulkar	2520	Rajkot	2007-02-11
Ishant	Sharma	2520	Rajkot	2007-02-11
Praveen	Kumar	2520	Rajkot	2007-02-11
MS	Dhoni	2520	Rajkot	2007-02-11
Yuvraj	Singh	2520	Rajkot	2007-02-11
Virender	Sehwag	2520	Rajkot	2007-02-11
Harbhajan	singh	2611	Southa...	2007-08-21
Saurav	Ganguly	2611	Southa...	2007-08-21
Rahul	Dravid	2611	Southa...	2007-08-21
Anil	Kumble	2611	Southa...	2007-08-21
Sachin	Tendulkar	2611	Southa...	2007-08-21
Ishant	Sharma	2611	Southa...	2007-08-21

Praveen	Kumar	2611	Southa...	2007-08-21
MS	Dhoni	2611	Southa...	2007-08-21
Yuvraj	Singh	2611	Southa...	2007-08-21
Virender	Sehwag	2611	Southa...	2007-08-21
Harbhajan	singh	2632	Mumbai	2007-10-17
Saurav	Ganguly	2632	Mumbai	2007-10-17
Rahul	Dravid	2632	Mumbai	2007-10-17
Anil	Kumble	2632	Mumbai	2007-10-17
Sachin	Tendulkar	2632	Mumbai	2007-10-17
Ishant	Sharma	2632	Mumbai	2007-10-17
Praveen	Kumar	2632	Mumbai	2007-10-17
MS	Dhoni	2632	Mumbai	2007-10-17
Yuvraj	Singh	2632	Mumbai	2007-10-17
Virender	Sehwag	2632	Mumbai	2007-10-17
Harbhajan	singh	2643	Guwahati	2007-11-05
Saurav	Ganguly	2643	Guwahati	2007-11-05
Rahul	Dravid	2643	Guwahati	2007-11-05
Anil	Kumble	2643	Guwahati	2007-11-05
Sachin	Tendulkar	2643	Guwahati	2007-11-05
Ishant	Sharma	2643	Guwahati	2007-11-05
Praveen	Kumar	2643	Guwahati	2007-11-05

MS	Dhoni	2643	Guwahati	2007-11-05
Yuvraj	Singh	2643	Guwahati	2007-11-05
Virender	Sehwag	2643	Guwahati	2007-11-05
Harbhajan	singh	2675	Melbourne	2008-02-10
Saurav	Ganguly	2675	Melbourne	2008-02-10
Rahul	Dravid	2675	Melbourne	2008-02-10
Anil	Kumble	2675	Melbourne	2008-02-10
Sachin	Tendulkar	2675	Melbourne	2008-02-10
Ishant	Sharma	2675	Melbourne	2008-02-10
Praveen	Kumar	2675	Melbourne	2008-02-10
MS	Dhoni	2675	Melbourne	2008-02-10
Yuvraj	Singh	2675	Melbourne	2008-02-10
Virender	Sehwag	2675	Melbourne	2008-02-10
Harbhajan	singh	2681	Adelaide	2008-02-19
Saurav	Ganguly	2681	Adelaide	2008-02-19
Rahul	Dravid	2681	Adelaide	2008-02-19
Anil	Kumble	2681	Adelaide	2008-02-19
Sachin	Tendulkar	2681	Adelaide	2008-02-19
Ishant	Sharma	2681	Adelaide	2008-02-19
Praveen	Kumar	2681	Adelaide	2008-02-19
MS	Dhoni	2681	Adelaide	2008-02-19

Yuvraj	Singh	2681	Adelaide	2008-02-19
Virender	Sehwag	2681	Adelaide	2008-02-19
Harbhajan	singh	2688	Sydney	2008-03-02
Saurav	Ganguly	2688	Sydney	2008-03-02
Rahul	Dravid	2688	Sydney	2008-03-02
Anil	Kumble	2688	Sydney	2008-03-02
Sachin	Tendulkar	2688	Sydney	2008-03-02
Ishant	Sharma	2688	Sydney	2008-03-02
Praveen	Kumar	2688	Sydney	2008-03-02
MS	Dhoni	2688	Sydney	2008-03-02
Yuvraj	Singh	2688	Sydney	2008-03-02
Virender	Sehwag	2688	Sydney	2008-03-02
Harbhajan	singh	2689	Brisbane	2008-03-04
Saurav	Ganguly	2689	Brisbane	2008-03-04
Rahul	Dravid	2689	Brisbane	2008-03-04
Anil	Kumble	2689	Brisbane	2008-03-04
Sachin	Tendulkar	2689	Brisbane	2008-03-04
Ishant	Sharma	2689	Brisbane	2008-03-04
Praveen	Kumar	2689	Brisbane	2008-03-04
MS	Dhoni	2689	Brisbane	2008-03-04
Yuvraj	Singh	2689	Brisbane	2008-03-04

Schema: d6_10

PlayerID	FName	LName	MatchID	Ground	Date
23001	Yuvraj	Singh	2750	Colombo	2008-08-24
21001	Virender	Sehwag	2750	Colombo	2008-08-24
406317	Sam	NULL	2755	Colombo	2008-08-24
98002	Harbhajan	Singh	2755	Colombo	2008-08-24
96003	Saurav	Ganguly	2755	Colombo	2008-08-24
96001	Rahul	Dravid	2755	Colombo	2008-08-24
90002	Anil	Kumble	2755	Colombo	2008-08-24
89001	Sachin	Tendulkar	2755	Colombo	2008-08-24
27002	Ishant	Sharma	2755	Colombo	2008-08-24
27001	Praveen	Kumar	2755	Colombo	2008-08-24
25001	MS	Dhoni	2755	Colombo	2008-08-24
23001	Yuvraj	Singh	2755	Colombo	2008-08-24
21001	Virender	Sehwag	2755	Colombo	2008-08-24

11 [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

```
SELECT B.PID AS PlayerID,
    M.Ground,
    SUM(B.NRuns) AS TotalRuns
FROM Batting B
INNER JOIN ODIMatch M ON B.MatchID = M.MatchID
WHERE M.Ground = 'Colombo'
GROUP BY B.PID, M.Ground;
```

Schema: d6_10

PlayerID	Ground	TotalRuns
23001	Colombo	0
25001	Colombo	71
91001	Colombo	60
94002	Colombo	17
92002	Colombo	1
27001	Colombo	2
98002	Colombo	2

12 [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

```
SELECT P.PlayerID, P.FName, P.LName,
    COALESCE(SUM(B.NRuns), 0) AS TotalCareerRuns
FROM Player P
LEFT JOIN Batting B ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName
ORDER BY TotalCareerRuns DESC;
```

Schema: d6_10

PlayerID	FName	LName	TotalCareerRuns
25001	MS	Dhoni	107
89001	Sachin	Tendulkar	91
91001	Sanath	Jayasuriya	60
24001	Andrew	Symonds	42
23001	Yuvraj	Singh	38
94002	Chaminda	Vaas	17
27001	Praveen	Kumar	9
99001	Breet	Lee	7
98002	Harbhajan	Singh	5
99002	Adam	Gilchrist	2
92002	Muthiah	Murlitharan	1
95001	Ricky	Ponting	1
21001	Virender	Sehwag	0

Schema: d6_10

PlayerID	FName	LName	TotalCareerRuns
90002	Anil	Kumble	0
92001	Shane	Warne	0
92003	Inzamam	UI-Huq	0
93002	Heath	Streak	0
93003	Gary	Kirsten	0
94004	Stephen	Fleming	0
95002	Shaun	Pollock	0
95003	Jacques	Kallis	0
96001	Rahul	Dravid	0
96002	Herschelle	Gibbs	0
96003	Saurav	Ganguly	0
97004	Daniel	Vettori	0
98001	Shahid	Afridi	0

13 [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

```
SELECT P.PlayerID, P.FName, P.LName, P.Country
FROM Player P
LEFT JOIN Bowling B ON P.PlayerID = B.PID
WHERE B.PID IS NULL;
```

Schema: d6_10

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa

Schema: d6_10

PlayerID	FName	LName	Country
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
300311	Michale	Clark	NULL
406317	Sam	NULL	India